APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 5)

SCENARIO: SUPPLIER ORDER LOG DATA CLEANING & AUDIT

ASSIGNMENT OVERVIEW: You are assisting the Procurement Lead at Apex Retail Group. The company receives raw supplier order logs containing order IDs, order dates, supplier names, unit costs, and order quantities.

Due to legacy database export issues, the log file contains duplicate entries, inconsistent supplier naming, missing pricing values, and text-formatted dates.

You must clean the dataset, document all transformations, handle missing values, and output a clean dataset summary for the Procurement Lead.

RAW SUPPLIER ORDER LOG DATASET:

```python
import pandas as pd
import numpy as np

supplier_log_data = {
    "order_id": ["SUP-901", "SUP-902", "SUP-903", "SUP-904", "SUP-904",
"SUP-905", "SUP-906", "SUP-907", "SUP-908", "SUP-909"],
    "order_date": ["2026-09-15", "2026-09-15", "2026-09-16", "2026-09-17",
"2026-09-17", "2026-09-18", "2026-09-18", "2026-09-19", "2026-09-20",
"2026-09-20"],
    "supplier_name": [" apex logistics ", "Global Freight", "APEX
LOGISTICS", "FastShip Inc", "FastShip Inc", " global freight ", "Apex
Logistics", "FastShip Inc", "GLOBAL FREIGHT", "FastShip Inc"],
    "unit_cost_usd": [120.00, 350.00, np.nan, 85.00, 85.00, 410.00, 115.00,
np.nan, 390.00, 90.00],
    "quantity_ordered": [50, 20, 40, 100, 100, 15, 60, 80, 25, 110]
}
```

PROBLEM 1: DEFECT AUDIT & DEDUPLICATION

In a Jupyter Notebook named `homework_session5_yourname.ipynb`:

1. Load `supplier_log_data` into DataFrame `df_supplier`.
2. Inspect initial row count (`.shape`) and report missing value counts per column.
3. Detect and drop exact duplicate records (`.drop_duplicates()`).
4. Display the resulting DataFrame shape after deduplication.

PROBLEM 2: STRING STANDARDIZATION, TYPE CASTING & IMPUTATION

1. Clean the `supplier_name` column: strip leading/trailing whitespace and convert text to Title Case.
2. Convert `order_date` to explicit `datetime64[ns]` using `pd.to_datetime()`.
3. Create a new calculated column `total_order_cost` equal to `unit_cost_usd * quantity_ordered`.
4. Calculate the median of `unit_cost_usd`. Fill missing values in `unit_cost_usd` with this median value.
5. Recalculate `total_order_cost` so that previously missing rows reflect the imputed unit cost.

PROBLEM 3: PROCUREMENT AUDIT BRIEFING

Write a 150-word formal briefing memo to the Procurement Lead containing:

1. Summary of data quality defects identified and corrected (duplicates removed, categories standardized).
2. Total spend calculation across all supplier orders after missing value imputation.
3. An explanation of how cleaning inconsistent supplier names prevents misrepresenting supplier order volumes.

SUBMISSION DEADLINE: Prior to Session 6.